UNIVERSITI SAINS MALAYSIA

School of Electrical and Electronics Engineering

Universiti Sains Malaysia, Engineering Campus

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EEM 441

Control & Instrumentation Laboratory

Experiment 9

INTERFACING FPGA TO AN ANALOG-TO-DIGITAL CONVERTER

Date : ______________________________

Group No. : ______________________________

Group Members : ______________________________ Matrix # ____________

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Experiment 9

Interfacing FPGA to Analog-to-Digital Converter

Objectives

1. To understand the operation of an analog-to-digital converter (ADC).
2. To interface an FPGA, via the Altera DE2 board, to an ADC.

Equipment / Software Required

1. Altera Quartus II software
   - version 9/10/12 for Cyclone II
   - version 9/10/12/latest for Cyclone IV
2. Altera DE2 board
3. Multimeter / oscilloscope
4. ADC0804 analog-to-digital converter
5. Resistors: $10\,\mathrm{k\Omega}$
6. Variable resistor (potentiometer), $10\,\mathrm{k\Omega}$
7. Capacitor: $150\,\mathrm{pF}$
8. 8 LEDs
9. 74LS245 octal bus transceiver

Introduction

This experiment explores interfacing an ADC chip to an FPGA. ADCs are among the most widely used data-acquisition devices: they translate an analog signal into a digital number that a processor (or, here, an FPGA) can read. A widely used ADC is the ADC0804 (National Semiconductor): it operates from $+5\,\mathrm{V}$ and has 8-bit resolution.

Besides resolution, *conversion time* is a major factor in judging an ADC: the time the ADC takes to convert its analog input to a digital number. For the ADC0804 this depends on the clock signal applied to the CLK R / CLK IN pins, but cannot be faster than $110\,\mu\mathrm{s}$.

The ADC0804's input range is set by the $V_{ref}/2$ pin, as shown in Table 9.1. The chip's outputs are 5 V logic; since the DE2 board's inputs only tolerate 3.3 V, an octal bus transceiver (74LS245) must be used to step the output voltage down before it reaches the FPGA.

$$D_{out} = \frac{V_{in}}{\text{step size}}$$

where D_{out} is the digital data output and V_{in} is the analog input voltage.

Data conversion with the ADC0804 follows this sequence (Figure 9.1):

i. Set $\overline{\text{CS}} = 0$ and send a low-to-high pulse on the $\overline{\text{WR}}$ pin to start the conversion.

ii. Monitor the $\overline{\text{INTR}}$ pin: while it is high, the conversion is still in progress; once it goes low, the conversion is complete.

iii. With $\overline{\text{INTR}}$ low, set $\overline{\text{CS}} = 0$ again and send a high-to-low pulse on the $\overline{\text{RD}}$ pin to read the data out of the ADC0804.

Table 9.1: $V_{ref}/2$ relation to V_{in} range.

$V_{ref}/2$ (V)	V_{in} range (V)	Step size (mV)
Not connected	0 to 5	$5/256 = 19.53$
2.0	0 to 4	$4/255 = 15.62$
1.5	0 to 3	$3/256 = 11.71$
1.28	0 to 2.56	$2.56/256 = 10.00$
1.0	0 to 2	$2/256 = 7.81$
0.5	0 to 1	$1/256 = 3.90$

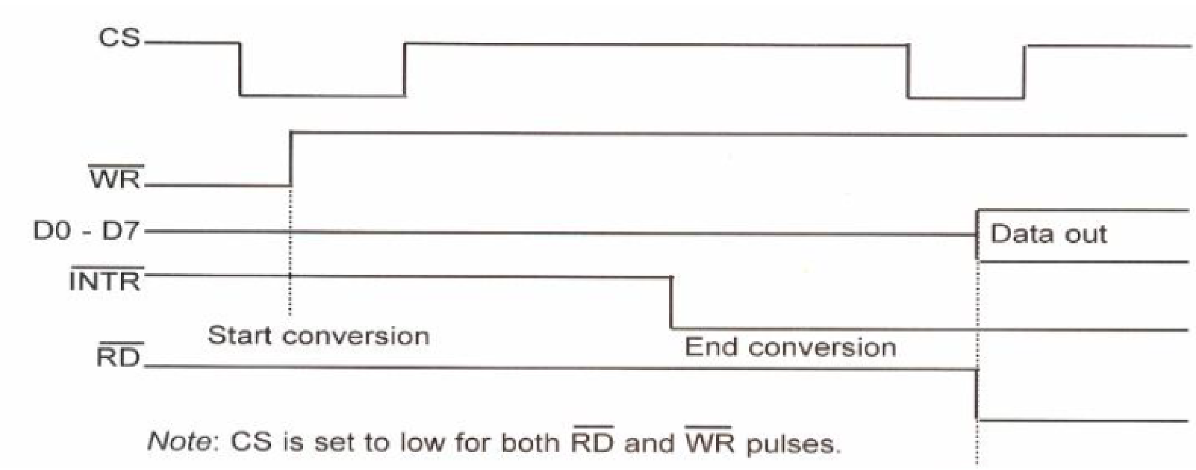

Figure 9.1: Read and write timing for the ADC0804.

Procedure

Activity 1 — Free-running mode

Objective: run the ADC0804 in free-running mode.

1. Connect the ADC0804 as shown in Figure 9.2 for free-running mode.

2. Vary the input signal from 0 to 5 V using the 10 kΩ potentiometer shown in Figure 9.2 as the ADC's input. The input must not exceed 5 V.
3. Tabulate your results in Table 9.2.
4. Compare your hardware result with the value calculated using Equation (1).

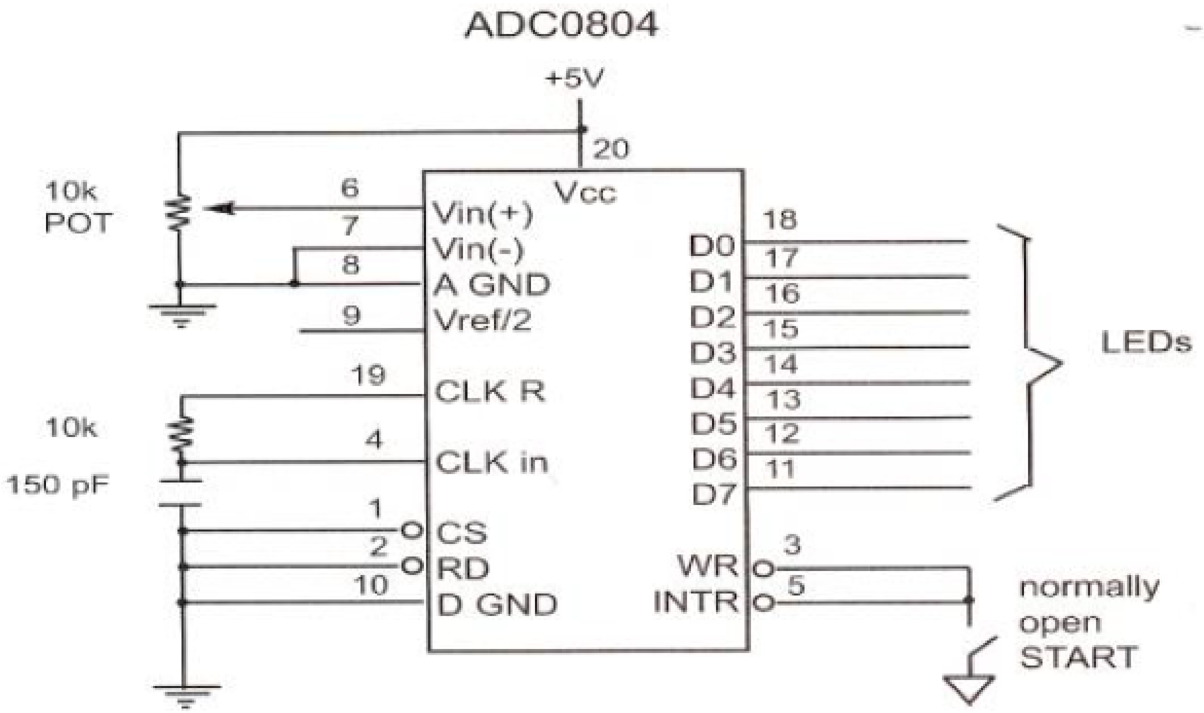

Figure 9.2: ADC0804 test circuit for free-running mode.

Table 9.2: Free-running mode output.

#	Input (V)	Output (hardware)	Output (calculation)
1	0.0		
2	0.5		
3	1.0		
4	1.5		
5	2.0		
6	2.5		
7	3.0		
8	3.5		
9	4.0		
10	4.5		

Activity 2 — Interfacing the ADC to the FPGA, displaying on LEDs

Objective: interface the ADC to the FPGA and display the digitised result on LEDs.

1. Use the circuit from Figure 9.2, modified as follows.
2. Connect the $\overline{\text{RD}}$ and $\overline{\text{WR}}$ pins to general-purpose I/O pins on the DE2 board, declared as **outputs**.
3. Connect the $\overline{\text{INTR}}$ and D0–D7 pins to the inputs of a 74LS245. Connect the 74LS245's outputs to general-purpose I/O pins on the DE2 board, declared as **inputs**. Refer to the DE2 user manual for pin locations.
4. In Verilog, generate the $\overline{\text{WR}}$ pulse to start conversion, poll $\overline{\text{INTR}}$ for end of conversion, then pulse $\overline{\text{RD}}$ to read the data on D0–D7.
5. Assign the FPGA's digitised outputs to the LED pins.
6. Compile, simulate, and download the program to the DE2 board.
7. Use the potentiometer to vary the input value.
8. Tabulate your results in Table 9.3.
9. Print and submit the Verilog code, simulation result, and experimental results.

Table 9.3: FPGA-interfaced ADC output.

#	Input (V)	Output
1	0.8	
2	1.7	
3	2.3	
4	2.8	
5	3.1	
6	3.6	
7	3.9	
8	4.1	
9	4.6	
10	4.8	

Activity 3 — Displaying the ADC result on the LCD

Objective: interface the ADC0804 to the DE2 board and display the result on the LCD.

1. Using the same circuit as in Activity 2, write Verilog code to read data from the ADC and display the result on the LCD (see Experiment 5 for the LCD interface).
2. Print and submit the Verilog code and simulation result.